Chapter 2 Limits
Tutorial

Computation of Limits

XJTLU | SCHOOL OF MATHEMATICS AND PHYSICS

Tutorial (Computation of Limits)

Methods for Evaluating Limits (before l'Hôpital)

- Direct Substitution 直接代入

  If $f(x)$ is continuous, c is in its domain, then $\lim_{x \to c} f(x) = f(c)$.

  Every elementary function is continuous in the interior of its domain.

- Handling Indeterminate Forms $0/0$
  - Factor the numerator and denominator to cancel the common factor of $(x - c)$. 约分
  - If radicals are involved, rationalize the expression to eliminate the zero factor. 有理化
  - Substitution of equivalent infinitesimals. 等价无穷小代换
  - Substitution of variables: 变量代换

    the substitution $h = x - c$ transforms $x \to c$ into $h \to 0$;
    the substitution $h = 1/x$ transforms $x \to \infty$ into $h \to 0^+$.

Methods for Evaluating Limits (continued)

- Handling Indeterminate Forms ∞/∞
  - Divide by the highest power of x in the denominator. 上下同除
  - When $x \to -\infty$, be careful with signs: $\sqrt{x^2} = |x|$.
- Handling Indeterminate Forms 1^∞
  - Two Important Limits (and their variants): $\lim\limits_{x\to 0} \frac{\sin x}{x} = \lim\limits_{x\to\pm\infty} x \sin\frac{1}{x} = 1$

$$\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n = \lim_{x\to\pm\infty}\left(1+\frac{1}{x}\right)^x = \lim_{h\to 0}(1+h)^{1/h} = e$$

  - $f^g = \left(1 + (f-1)\right)^g$ 凑
  - $f^g = e^{\ln(f^g)} = e^{g\ln f}$
- Squeeze Theorem
  - infinitesimal × bounded = infinitesimal 无穷小 × 有界 = 无穷小
  - $\lim\limits_{x\to 0} x\sin\frac{1}{x} = \lim\limits_{x\to 0} x\cos\frac{1}{x} = \lim\limits_{x\to\pm\infty}\frac{\sin x}{x} = \lim\limits_{x\to\pm\infty}\frac{\cos x}{x} = 0$

$\lim\limits_{x\to 0} \underbrace{x^2}_{\to 0} \underbrace{\sin\frac{1}{x}}_{\text{bounded}} = 0$,

or, you can write

$\lim\limits_{x\to 0} x^2 \sin\frac{1}{x} = \lim\limits_{x\to 0} x\cdot\left(x\sin\frac{1}{x}\right) = 0\cdot 0 = 0.$

Warnings

- The value for x must be substituted into all parts of the expression at the same time. A common mistake is to perform a partial substitution.
变量的代入必须一步到位，覆盖整个表达式. 不能分步代入或者只代入一部分.

Example

$$\lim_{x\to 0}\frac{\sin x}{x} \neq \lim_{x\to 0}\frac{\sin 0}{x} = \lim_{x\to 0}\frac{0}{x} = 0;$$

$$\lim_{x\to\infty}\left(\frac{\left(1+\frac{1}{x}\right)^x}{e}\right)^x \neq \lim_{x\to\infty}\left(\frac{e}{e}\right)^x = \lim_{x\to\infty} 1^x = 1.$$

- Substitution of equivalent infinitesimals can only be applied to terms that are part of a multiplication or division. 等价无穷小只能代换乘除的情况.
It CANNOT be used on terms within a sum or difference.
如果分子或分母为若干项之和，那么不能对其中的某个加项作等价无穷小的代换.

Example

$$\lim_{x\to 0}\frac{\tan x-\sin x}{x^3} \neq \lim_{x\to 0}\frac{x-x}{x^3} = 0.$$

$$\lim_{x\to\infty}\left(\frac{\left(1+\frac{1}{x}\right)^x}{e}\right)^x = \lim_{x\to\infty}\left(\frac{e^{x\ln\left(1+\frac{1}{x}\right)}}{e}\right)^x \neq \lim_{x\to\infty}\left(\frac{e^{x\cdot\frac{1}{x}}}{e}\right)^x = \lim_{x\to\infty}\left(\frac{e}{e}\right)^x = 1.$$

Exercise 1 Use substitution by equivalent infinitesimals to compute the following limits.

(a) $\lim\limits_{x\to 0}\dfrac{\tan x^3}{\sin^2 x}$ (b) $\lim\limits_{x\to 0}\dfrac{1-\cos x}{\sqrt{1-x^2}-1}$

Solution

(a) As $x\to 0$, $\tan x^3 \sim x^3$,

$(\sin x)^2 \sim x^2$.

$$\lim_{x\to 0}\frac{\tan x^3}{\sin^2 x} = \lim_{x\to 0}\frac{x^3}{x^2} = \lim_{x\to 0} x = 0.$$

(b) As $x\to 0$, $1-\cos x \sim \frac{1}{2}x^2$

$\sqrt{1+x}-1 \sim \frac{1}{2}x$ $\boxed{\text{Recall } (1+x)^\alpha - 1 \sim \alpha x,\ x\to 0}$

so $\sqrt{1-x^2}-1 \sim \frac{1}{2}\cdot(-x^2)$.

$$\lim_{x\to 0}\frac{1-\cos x}{\sqrt{1-x^2}-1} = \lim_{x\to 0}\frac{\frac{1}{2}x^2}{-\frac{1}{2}x^2} = -1.$$

Exercise 2 Find the following limits.

(a) $\displaystyle\lim_{x\to 1}\frac{\arcsin(1-x)}{\ln x}$ (b) $\displaystyle\lim_{x\to 0}\frac{\sin 2x}{\sqrt{x+2}-\sqrt{2}}$

Solution

(a) Let $h = x-1$. Then $h\to 0$ as $x\to 1$.

$$\lim_{x\to 1}\frac{\arcsin(1-x)}{\ln x} = \lim_{h\to 0}\frac{\arcsin(-h)}{\ln(h+1)} = \lim_{h\to 0}\frac{-h}{h} = -1$$

substitution of equivalent infinitesimal.

$$\boxed{\begin{array}{ll}\arcsin x \sim x & \text{as } x\to 0\\ \ln(1+x)\sim x & \end{array}}$$

(b) By rationalization.

$$\lim_{x\to 0}\frac{\sin 2x}{\sqrt{x+2}-\sqrt{2}} = \lim_{x\to 0}\frac{\sin 2x}{\sqrt{x+2}-\sqrt{2}}\cdot\frac{\sqrt{x+2}+\sqrt{2}}{\sqrt{x+2}+\sqrt{2}}$$

$$= \lim_{x\to 0}\frac{\sin 2x}{x}\cdot\left(\sqrt{x+2}+\sqrt{2}\right) = 2\left(\sqrt{2}+\sqrt{2}\right) = 4\sqrt{2}.$$

By substitution of equivalent infinitesimal:

As $x\to 0$, $\sin 2x \sim 2x$,

$$\sqrt{x+2}-\sqrt{2} = \sqrt{2}\left(\sqrt{1+\frac{x}{2}}-1\right) \sim \sqrt{2}\cdot\frac{1}{2}\left(\frac{x}{2}\right) = \frac{\sqrt{2}}{4}x$$

So $\displaystyle\lim_{x\to 0}\frac{\sin 2x}{\sqrt{x+2}-\sqrt{2}} = \lim_{x\to 0}\frac{2x}{\frac{\sqrt{2}}{4}x} = 4\sqrt{2}.$

Exercise 3 Find the following limits.

(a) $\lim_{x\to\infty}\left(\frac{x+2}{x-2}\right)^{x}$ $\quad$ (b) $\lim_{x\to 0}(\cos x - \sin x)^{1/x}$ $\quad$ (c) $\lim_{x\to\pi/2}(\sin x)^{\cos x}$

(b): 1^{∞} $\quad$ (c): 1^{0}

Solution

(a) $\lim_{x\to\infty}\left(\frac{x+2}{x-2}\right)^{x} = \lim_{x\to\infty}\left(\frac{x-2+4}{x-2}\right)^{x} = \lim_{x\to\infty}\underbrace{\left(1+\frac{4}{x-2}\right)^{\frac{x-2}{4}}}_{\to e}{}^{\cdot\frac{4x}{x-2}} \overset{\lim_{x\to\infty}\frac{4}{x-2}=0}{\underset{\lim_{h\to 0}(1+h)^{1/h}=e}{=\!=}} e^{\lim_{x\to\infty}\frac{4x}{x-2}} = e^{\lim_{x\to\infty}\frac{4}{1-\frac{2}{x}}} = e^{4}.$

(b) $\lim_{x\to 0}(\cos x - \sin x)^{1/x} = \lim_{x\to 0}\left(1+\underline{\cos x - \sin x - 1}\right)^{1/x} = \lim_{x\to 0}\left(1+\underline{\cos x - \sin x - 1}\right)^{\frac{1}{\cos x - \sin x - 1}\cdot\frac{\cos x - \sin x - 1}{x}}$

$= e^{\lim_{x\to 0}\frac{\cos x - \sin x - 1}{x}} = e^{\lim_{x\to 0}\frac{\cos x - 1}{x} - \lim_{x\to 0}\frac{\sin x}{x}} \overset{1-\cos x\sim\frac{1}{2}x^{2}}{\underset{\text{as } x\to 0}{=\!=}} e^{0-1} = \frac{1}{e}.$

(c) $\lim_{x\to\frac{\pi}{2}}\sin x = 1$, $\lim_{x\to\frac{\pi}{2}}\cos x = 0$. So $\lim_{x\to\frac{\pi}{2}}(\sin x)^{\cos x} = 1^{0} = 1.$

Exercise 4 Find the limit $\lim_{x\to\frac{\pi}{2}}(1+\cos x)^{\frac{1}{2x-\pi}}$.

$(1+0)^{\infty}$

Solution

Let $h = x-\frac{\pi}{2}$. Then $x = h+\frac{\pi}{2}$, $\cos x = \cos\left(h+\frac{\pi}{2}\right) \overset{\cos(\frac{\pi}{2}-\alpha)=\sin\alpha}{=\!=\!=} -\sin h$.

$$\lim_{x\to\frac{\pi}{2}}(1+\cos x)^{\frac{1}{2x-\pi}} = \lim_{h\to0}(1-\sin h)^{1/2h} = \lim_{h\to0}\left(1+(-\sin h)\right)^{\frac{1}{-\sin h}\cdot\frac{-\sin h}{2h}} = e^{-\frac{1}{2}}.$$

Another method:

$$\lim_{x\to\frac{\pi}{2}}(1+\cos x)^{\frac{1}{2x-\pi}} = \lim_{x\to\frac{\pi}{2}}(1+\underset{\to 0}{\boxed{\cos x}})^{\frac{1}{\cos x}\cdot\frac{\cos x}{2x-\pi}}$$

$$= \lim_{x\to\frac{\pi}{2}}\Big(\underbrace{(1+\cos x)^{\frac{1}{\cos x}}}_{\to e}\Big)^{\frac{\cos x}{2x-\pi}} = e^{-\frac{1}{2}}.$$

Here $\lim_{x\to\frac{\pi}{2}}\frac{\cos x}{2x-\pi} \overset{\text{Let } h=x-\frac{\pi}{2}}{=\!=\!=} \lim_{h\to0}\frac{\cos(h+\frac{\pi}{2})}{2(h+\frac{\pi}{2})-\pi} \overset{\cos(\frac{\pi}{2}-\alpha)=\sin\alpha}{=\!=\!=} \lim_{h\to0}\frac{\sin(-h)}{2h} = \lim_{h\to0}\frac{-1}{2}\frac{\sin h}{h} = -\frac{1}{2}$.

Recall If $\lim_{x\to c} f(x) = A > 0$, $\lim_{x\to c} g(x) = B$, then

$$\lim_{x\to c}(f(x))^{g(x)} = \left(\lim_{x\to c} f(x)\right)^{\lim_{x\to c} g(x)} = A^B.$$

Some Advice for Learning Calculus

Passive Learning → Active Learning

- **Learn before class** —— Do not just “read” the textbook.
  - If you're hearing of a concept for the first time in class,
    it indicates a flawed learning method.
  - Try to **do some exercises** during the pre-class preparation.
    Pinpoint the content that demands focused attention during the class.
- **Learn during class** —— Know how to **take notes**.
- **Learn after class** —— Paper assignments are not enough.
  - I know at least 10% of students can **finish all the textbook exercises**. You can too.
  - I also know that at least 10% of students have never **attended office hours**.
    Don't be one of them. I am here to help you.

If You Want to Learn More

- **Change Learning Purpose**
  It's NOT all about exams and competitions.
  – Know what you want to do, so you know what to learn.
  – Enjoy life in college.
- **AI**
- **LaTeX & Markdown**
- **If you want to learn more math**
  – Mathematical Modeling
  – Advanced Linear Algebra
  – Mathematical Analysis
  – Abstract Algebra